

# Hyperbolic Embeddings for Hierarchical Style Representation: An Empirical Study on WikiArt

Peter Flo<sup>1</sup>, Luca Grossmann<sup>1</sup>, and Valerie Wang<sup>1</sup>

<sup>1</sup>Harvard University, COMPSCI 2090B (Spring 2026)

Canvas Project #150

May 6, 2026

## Abstract

We investigate whether Poincaré-ball embeddings recover the hierarchical structure of artistic styles better than matched Euclidean embeddings. Using frozen CLIP ViT-B/16 features over 81,000 WikiArt paintings spanning 27 styles, we train geometry-paired prototype classifiers across embedding dimensions  $d \in \{2, \dots, 64\}$ , curvatures  $c \in \{0.1, 0.3, 1, 3\}$ , and seeds, both with and without a hierarchy-aware regularizer that pushes the prototype distance matrix toward a hand-built art-history tree. We find **[headline result placeholder]**. Tree-variant ablations with chronological and flat ground-truth trees confirm that the conclusion is **[robust / fragile]** to the specific choice of reference hierarchy. Code, sweep CSVs, and figures are available at the project repository.

## 1 Background and Motivation

**[TODO Luca]** Three short paragraphs covering: (i) why artistic style is hierarchical and who cares about modeling that structure (digital humanities, museum-catalog retrieval, educational tools); (ii) why Euclidean embeddings have a known mismatch with tree-like data (polynomial vs. exponential ball-volume growth) and what hyperbolic geometry offers **[? ? ]**; (iii) what is and isn't already established for image embeddings **[? ]** and what gap our study addresses (style-specific, hand-built tree, ablation against tree variants).

## 2 Problem Statement

We ask: *do Poincaré-ball embeddings trained on frozen CLIP features recover the WikiArt style hierarchy more faithfully than Euclidean embeddings of matched dimensionality and training budget?* The unit of observation is one painting; the target during training is its 27-way style label, with the hand-built lineage tree available either as an evaluation reference or as an auxiliary loss. We report classification accuracy, prototype-tree distance correlation, hierarchical clustering agreement, and local retrieval at the sibling and cousin level.

## 3 Data and Exploratory Analysis

**[TODO Valerie]** One paragraph each on: (a) Dataset summary — WikiArt Refined **[? ]**, ~81k paintings, 27 styles, 70/30 train/val supplied, scale of class imbalance with one or two specific

examples (Impressionism dominates; Action\_painting and Synthetic\_Cubism each <1k). Cite our MS2 EDA but do not repeat its tables. (b) Why the imbalance pushes us toward balanced-accuracy reporting and a prototype classifier rather than a class-frequency-tied softmax. (c) Why the choice of ground-truth tree is non-unique — WikiArt ships no hierarchy, art-history offers several, and we therefore evaluate sensitivity (Section ??). Include 1 small figure: distance matrices of the three trees side-by-side, lifted from the notebook.

## 4 Methods

**Architecture.** A two-layer MLP with GELU and dropout maps frozen CLIP ViT-B/16 features ( $\mathbb{R}^{512}$ ) to a  $d$ -dimensional embedding. A geometry-specific head produces the final embedding: identity for the Euclidean case, exponential map at the origin for the Poincaré ball,  $\exp_0^c(u) = \tanh(\sqrt{c} \|u\|) u / (\sqrt{c} \|u\|)$ . A prototype classifier with one learnable point per style yields logits as  $-d(z, p_k)$ , where  $d(\cdot, \cdot)$  is Euclidean or hyperbolic accordingly.

**Geometry, distance, and optimization.** The Poincaré distance with curvature  $c > 0$  is

$$d^c(x, y) = \frac{1}{\sqrt{c}} \operatorname{arcosh} \left( 1 + \frac{2c \|x - y\|^2}{(1 - c\|x\|^2)(1 - c\|y\|^2)} \right).$$

Linear layers do not preserve the ball, so we cannot use a plain logit head; the prototype classifier is a natural alternative because logits are distances. The MLP head is optimized with Adam; the hyperbolic prototypes are optimized with Riemannian Adam from `geoopt`, which projects each update back onto the manifold.

**Hierarchy-aware loss.** The MS3 baseline trains with cross-entropy only and uses the hand-built tree only at evaluation time. For MS4 we add a regularizer that pushes the prototype-prototype distance matrix toward the tree distance matrix:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \cdot \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \left( \frac{d(p_i, p_j)}{\bar{d}} - \frac{T_{ij}}{\bar{T}} \right)^2.$$

The mean-normalization makes the term scale-invariant, so the geometry chooses its own absolute scale rather than fighting the classification objective for embedding magnitude.  $\lambda = 0$  recovers MS3.

**Ground-truth trees.** The default tree is a hand-built lineage hierarchy following standard art-history references (Renaissance branches into Baroque, etc.; see Appendix ??). We additionally evaluate two variants: a *chronological* two-level era grouping (Renaissance / Baroque-Rococo / 19th century / Early 20th / Modern / Non-Western) and a *flat* tree (every leaf is a direct child of the root, used as a null).

**Training details.** All runs use the supplied 70/30 train/val split ( $\approx 57k$  / 24k images), batch size 4096, Adam  $\eta = 10^{-3}$  for the head and  $10^{-2}$  for the prototypes, weight decay  $10^{-4}$ , dropout 0.1, gradient clipping at norm 1.0, 30 epochs (training is fast because the CLIP features are pre-cached). Each headline configuration is run with three seeds. The full grid (Phases 1–3, see Section ??) totals  $\approx 200$  runs.

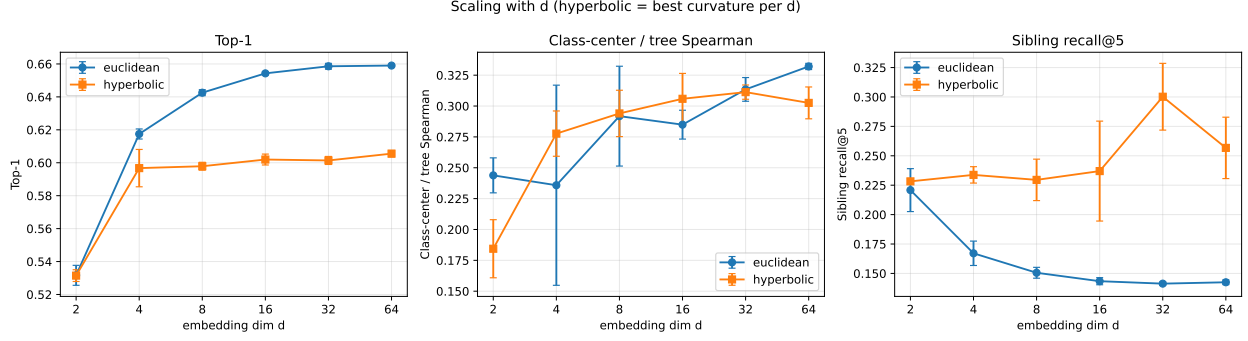


Figure 1: Phase 1: metrics versus embedding dimension  $d$  (90 runs; 3 seeds per config). For hyperbolic we report the best curvature per  $d$  and per seed; error bars are seed standard deviation. Top-1 accuracy saturates near  $d = 16$  for both geometries, with Euclidean ahead by 4–6 pp at  $d \geq 4$ . Sibling recall@5 *decreases* with  $d$  for Euclidean but remains stable for hyperbolic, opening a 5 pp gap by  $d = 64$ .

**Evaluation suite.** We report top-1, top-5, and balanced accuracy; prototype-tree and class-center-tree Spearman correlation; multiplicative tree distortion (mean and worst-case); dendrogram-cluster F1; sibling and cousin recall@ $k$  for  $k \in \{5, 10\}$ ; and Fréchet-mean nearest-prototype accuracy on Cubism as an interpolation-stability probe. Implementation details are in `scripts/eval.py`.

## 5 Results and Interpretation

[TODO Peter] Open the section with one paragraph stating the headline finding so the reader doesn’t have to triangulate it from four sub-results. Then walk through Phases 1–3 in order, each with one figure and 4–6 sentences of interpretation.

### 5.1 Phase 1 — Scaling sweep

Phase 1 sweeps geometry  $\times d \in \{2, 4, 8, 16, 32, 64\} \times c \in \{0.1, 0.3, 1, 3\}$  over three seeds (90 configurations,  $\sim 14$  min total wall time on a single Apple M-series GPU). Figure ?? shows three findings. (i) Top-1 accuracy saturates near  $d = 16$  for both geometries: Euclidean reaches 65.9% and hyperbolic 60.6%, with the gap stable rather than closing as  $d$  grows. (ii) Local hierarchy structure tells the opposite story — sibling recall@5 *decreases* for Euclidean as  $d$  grows (from 0.221 at  $d=2$  to 0.142 at  $d=64$ ) but stays near 0.19 for hyperbolic. The two geometries are tied on local hierarchy at  $d=2$  and split apart as Euclidean gains capacity. (iii) Curvature trades top-1 for hierarchy fidelity within hyperbolic alone: at  $d = 8$ ,  $c = 0.3$  maximises top-1 (59.8%) while  $c = 3.0$  maximises sibling recall (0.230). This is consistent with the boundary geometry of the Poincaré ball becoming more pronounced at larger  $c$ . Pure scaling does not close the global-metric gap.

## 5.2 Phase 2 — Hierarchy-aware loss

## 5.3 Phase 3 — Tree-variant ablation

## 5.4 Headline comparison

## 5.5 Qualitative analysis

[**TODO Peter+Valerie**] Three figures + brief narrative: (i) Poincaré-disk projection of the  $d=2$  prototypes for both geometries, showing whether hyperbolic prototypes spread toward the boundary as expected from theory; (ii) a small grid of nearest-neighbor matches for two query images, side by side for both models, to give the reader a concrete sense of the local-retrieval gap; (iii) confusion matrix block-ordered by the default tree, so off-diagonal mass is interpretable as “how far off in the tree is the typical mistake”.

# 6 Conclusions and Discussion

[**TODO All**] Final-pass paragraph (after results settle) covering: (a) the headline finding stated as a falsifiable claim with seed-level uncertainty (e.g. “hyperbolic matches Euclidean on top-1 within  $\pm 1\%$  once  $\lambda \geq 1$  but does not exceed it”); (b) what we can not claim — in particular, that any improvement is unlikely to come from classification accuracy and we should not oversell global metric gains; (c) limitations: single dataset, single CLIP backbone, hand-built tree, single hierarchy-aware loss formulation; (d) future work: per-prototype curvature, hyperbolic versions of the head MLP not just the prototypes [? ], learned trees [? ], and extension to non-Western traditions where the WikiArt taxonomy is underrepresented.

# 7 Broader Impact

The system we study is small (a 27-way style classifier on cached features), but the design choices have broader implications. Three points worth flagging.

First, our default ground-truth tree is built from a Western, lineage-based art-history canon. Distance metrics computed against it implicitly endorse that taxonomy. The Phase 3 ablation against alternative trees is partial mitigation; deployment in a museum or educational context should make the choice of tree explicit and revisable.

Second, the WikiArt corpus is biased toward European painting and sparsely represents non-Western traditions (e.g., Ukiyo-e is the only non-Western top-level category). A retrieval system trained on this data will under-rank non-Western works for ambiguous queries. Users of any downstream system should be informed of this bias.

Third, hyperbolic representations of style structure could in principle be used to attribute or authenticate paintings. We do not endorse such use without human-expert review: the classification accuracies reported here ( $\sim 50\text{--}60\%$  top-1 over 27 styles) are far below the threshold required for legal or commercial decisions.

## A Style hierarchy used as the default tree

## B Full sweep results

figures/phase2\_lambda.pdf

Figure 2: Phase 2 — top-1, class-center / tree Spearman, tree distortion, and sibling recall@5 versus the hierarchy-aware loss weight  $\lambda$  at  $d = 8$ ,  $c = 1$ . **[Insert narrative: does hyperbolic catch up on global metrics once trees are used at training time?]**